

3

Phase Portraits for Planar Systems

Given the Linearity Principle from the previous chapter, we may now compute the general solution of any planar system. There is a seemingly endless number of distinct cases, but we will see that these represent in the simplest possible form nearly all of the types of solutions we will encounter in the higher-dimensional case.

3.1 Real Distinct Eigenvalues

Consider $X' = AX$ and suppose that A has two real eigenvalues $\lambda_1 < \lambda_2$. Assuming for the moment that $\lambda_i \neq 0$, there are three cases to consider:

1. $\lambda_1 < 0 < \lambda_2$
2. $\lambda_1 < \lambda_2 < 0$
3. $0 < \lambda_1 < \lambda_2$

We give a specific example of each case; any system that falls into any one of these three categories may be handled similarly, as we show later.

Example. (Saddle) First consider the simple system $X' = AX$, where

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

with $\lambda_1 < 0 < \lambda_2$. This can be solved immediately since the system decouples into two unrelated first-order equations:

$$\begin{aligned} x' &= \lambda_1 x \\ y' &= \lambda_2 y. \end{aligned}$$

We already know how to solve these equations, but, having in mind what comes later, let's find the eigenvalues and eigenvectors. The characteristic equation is

$$(\lambda - \lambda_1)(\lambda - \lambda_2) = 0,$$

so λ_1 and λ_2 are the eigenvalues. An eigenvector corresponding to λ_1 is $(1, 0)$ and to λ_2 is $(0, 1)$. Thus, we find the general solution

$$X(t) = \alpha e^{\lambda_1 t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta e^{\lambda_2 t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Since $\lambda_1 < 0$, the straight-line solutions of the form $\alpha e^{\lambda_1 t}(1, 0)$ lie on the x -axis and tend to $(0, 0)$ as $t \rightarrow \infty$. This axis is called the *stable line*. Since $\lambda_2 > 0$, the solutions $\beta e^{\lambda_2 t}(0, 1)$ lie on the y -axis and tend away from $(0, 0)$ as $t \rightarrow \infty$; this axis is the *unstable line*. All other solutions (with $\alpha, \beta \neq 0$) tend to ∞ in the direction of the unstable line, as $t \rightarrow \infty$, since $X(t)$ comes closer and closer to $(0, \beta e^{\lambda_2 t})$ as t increases. In backward time, these solutions tend to ∞ in the direction of the stable line. ■

In Figure 3.1 we have plotted the *phase portrait* of this system. The phase portrait is a picture of a collection of representative solution curves of the system in $\mathbb{R}^2$, which we call the *phase plane*. The equilibrium point of a system of this type (eigenvalues satisfying $\lambda_1 < 0 < \lambda_2$) is called a *saddle*.

For a slightly more complicated example of this type, consider $X' = AX$, where

$$A = \begin{pmatrix} 1 & 3 \\ 1 & -1 \end{pmatrix}.$$

As we saw in Chapter 2, the eigenvalues of A are ± 2 . The eigenvector associated with $\lambda = 2$ is the vector $(3, 1)$; the eigenvector associated with $\lambda = -2$ is

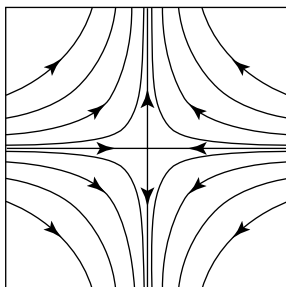


Figure 3.1 Saddle phase portrait for $x' = -x$, $y' = y$.

$(1, -1)$. Thus, we have an unstable line that contains straight-line solutions of the form

$$X_1(t) = \alpha e^{2t} \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

each of which tends away from the origin as $t \rightarrow \infty$. The stable line contains the straight-line solutions

$$X_2(t) = \beta e^{-2t} \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

which tend toward the origin as $t \rightarrow \infty$. By the Linearity Principle, any other solution assumes the form

$$X(t) = \alpha e^{2t} \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \beta e^{-2t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

for some α, β . Note that, if $\alpha \neq 0$, as $t \rightarrow \infty$, we have

$$X(t) \sim \alpha e^{2t} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = X_1(t),$$

whereas, if $\beta \neq 0$, as $t \rightarrow -\infty$,

$$X(t) \sim \beta e^{-2t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = X_2(t).$$

Thus, as time increases, the typical solution approaches $X_1(t)$ while, as time decreases, this solution tends toward $X_2(t)$, just as in the previous case. Figure 3.2 displays this phase portrait.

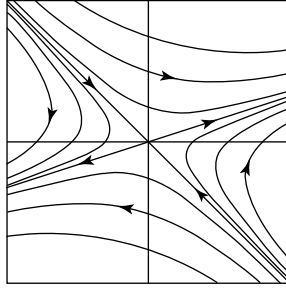


Figure 3.2 Saddle
phase portrait for
 $x' = x + 3y$, $y' = x - y$.

In the general case where A has a positive and negative eigenvalue, we always find a similar stable and unstable line on which solutions tend toward or away from the origin. All other solutions approach the unstable line as $t \rightarrow \infty$, and tend toward the stable line as $t \rightarrow -\infty$.

Example. (Sink) Now consider the case $X' = AX$ where

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

but $\lambda_1 < \lambda_2 < 0$. As before, we find two straight-line solutions and then the general solution

$$X(t) = \alpha e^{\lambda_1 t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta e^{\lambda_2 t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Unlike the saddle case, now all solutions tend to $(0,0)$ as $t \rightarrow \infty$. The question is this: How do they approach the origin? To answer this, we compute the slope dy/dx of a solution with $\beta \neq 0$. We write

$$\begin{aligned} x(t) &= \alpha e^{\lambda_1 t} \\ y(t) &= \beta e^{\lambda_2 t} \end{aligned}$$

and compute

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\lambda_2 \beta e^{\lambda_2 t}}{\lambda_1 \alpha e^{\lambda_1 t}} = \frac{\lambda_2 \beta}{\lambda_1 \alpha} e^{(\lambda_2 - \lambda_1)t}.$$

Since $\lambda_2 - \lambda_1 > 0$, it follows that these slopes approach $\pm\infty$ (provided $\beta \neq 0$). Thus these solutions tend to the origin tangentially to the y -axis. ■

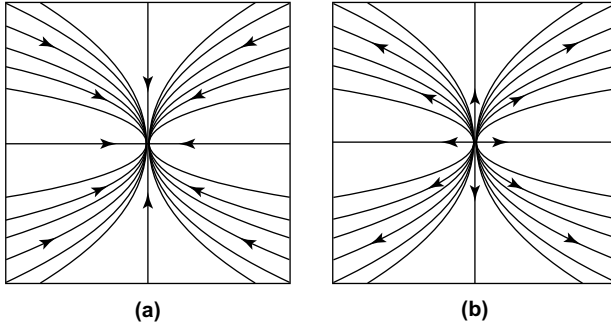


Figure 3.3 Phase portraits for a sink and a source.

Since $\lambda_1 < \lambda_2 < 0$, we call λ_1 the stronger eigenvalue and λ_2 the weaker eigenvalue. The reason for this in this particular case is that the x -coordinates of solutions tend to 0 much more quickly than the y -coordinates. This accounts for why solutions (except those on the line corresponding to λ_1 -eigenvector) tend to “hug” the straight-line solution corresponding to the weaker eigenvalue as they approach the origin. The phase portrait for this system is displayed in Figure 3.3a. In this case the equilibrium point is called a *sink*.

More generally, if the system has eigenvalues $\lambda_1 < \lambda_2 < 0$ with eigenvectors (u_1, u_2) and (v_1, v_2) respectively, then the general solution is

$$\alpha e^{\lambda_1 t} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \beta e^{\lambda_2 t} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

The slope of this solution is given by

$$\begin{aligned} \frac{dy}{dx} &= \frac{\lambda_1 \alpha e^{\lambda_1 t} u_2 + \lambda_2 \beta e^{\lambda_2 t} v_2}{\lambda_1 \alpha e^{\lambda_1 t} u_1 + \lambda_2 \beta e^{\lambda_2 t} v_1} \\ &= \left(\frac{\lambda_1 \alpha e^{\lambda_1 t} u_2 + \lambda_2 \beta e^{\lambda_2 t} v_2}{\lambda_1 \alpha e^{\lambda_1 t} u_1 + \lambda_2 \beta e^{\lambda_2 t} v_1} \right) \frac{e^{-\lambda_2 t}}{e^{-\lambda_2 t}} \\ &= \frac{\lambda_1 \alpha e^{(\lambda_1 - \lambda_2)t} u_2 + \lambda_2 \beta v_2}{\lambda_1 \alpha e^{(\lambda_1 - \lambda_2)t} u_1 + \lambda_2 \beta v_1}, \end{aligned}$$

which tends to the slope v_2/v_1 of the λ_2 -eigenvector, unless we have $\beta = 0$. If $\beta = 0$, our solution is the straight-line solution corresponding to the eigenvalue λ_1 . Thus, in this case as well, all solutions (except those on the straight line corresponding to the stronger eigenvalue) tend to the origin tangentially to the straight-line solution corresponding to the weaker eigenvalue.

Example. (Source) When the matrix

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

satisfies $0 < \lambda_2 < \lambda_1$, our vector field may be regarded as the negative of the previous example. The general solution and phase portrait remain the same, except that all solutions now tend away from $(0,0)$ along the same paths. See Figure 3.3b. ■

One may argue that we are presenting examples here that are much too simple. Although this is true, we will soon see that any system of differential equations with a matrix that has real distinct eigenvalues can be manipulated into the preceding special forms by changing coordinates.

Finally, a special case occurs if one of the eigenvalues is equal to 0. As we have seen, there is a straight line of equilibrium points in this case. If the other eigenvalue λ is nonzero, then the sign of λ determines whether the other solutions tend toward or away from these equilibria (see Exercises 10 and 11 of this chapter).

3.2 Complex Eigenvalues

It may happen that the roots of the characteristic polynomial are complex numbers. In analogy with the real case, we call these roots *complex eigenvalues*. When the matrix A has complex eigenvalues, we no longer have straight-line solutions. However, we can still derive the general solution as before by using a few tricks involving complex numbers and functions. The following examples indicate the general procedure.

Example. (Center) Consider $X' = AX$ with

$$A = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}$$

and $\beta \neq 0$. The characteristic polynomial is $\lambda^2 + \beta^2 = 0$, so the eigenvalues are now the imaginary numbers $\pm i\beta$. Without worrying about the resulting complex vectors, we react just as before to find the eigenvector corresponding to $\lambda = i\beta$. We therefore solve

$$\begin{pmatrix} -i\beta & \beta \\ -\beta & -i\beta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

or $i\beta x = \beta y$, since the second equation is redundant. Thus we find a complex eigenvector $(1, i)$, and so the function

$$X(t) = e^{i\beta t} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

is a complex solution of $X' = AX$.

Now in general it is not polite to hand someone a complex solution to a real system of differential equations, but we can remedy this with the help of Euler's formula:

$$e^{i\beta t} = \cos \beta t + i \sin \beta t.$$

Using this fact, we rewrite the solution as

$$X(t) = \begin{pmatrix} \cos \beta t + i \sin \beta t \\ i(\cos \beta t + i \sin \beta t) \end{pmatrix} = \begin{pmatrix} \cos \beta t + i \sin \beta t \\ -\sin \beta t + i \cos \beta t \end{pmatrix}.$$

Better yet, by breaking $X(t)$ into its real and imaginary parts, we have

$$X(t) = X_{re}(t) + iX_{im}(t),$$

where

$$X_{re}(t) = \begin{pmatrix} \cos \beta t \\ -\sin \beta t \end{pmatrix}, \quad X_{im}(t) = \begin{pmatrix} \sin \beta t \\ \cos \beta t \end{pmatrix}.$$

But now we see that both $X_{re}(t)$ and $X_{im}(t)$ are (real!) solutions of the original system. To see this, we simply check

$$\begin{aligned} X'_{re}(t) + iX'_{im}(t) &= X'(t) \\ &= AX(t) \\ &= A(X_{re}(t) + iX_{im}(t)) \\ &= AX_{re} + iAX_{im}(t). \end{aligned}$$

Equating the real and imaginary parts of this equation yields $X'_{re} = AX_{re}$ and $X'_{im} = AX_{im}$, which shows that both are indeed solutions. Moreover, since

$$X_{re}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad X_{im}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

the linear combination of these solutions,

$$X(t) = c_1 X_{re}(t) + c_2 X_{im}(t),$$

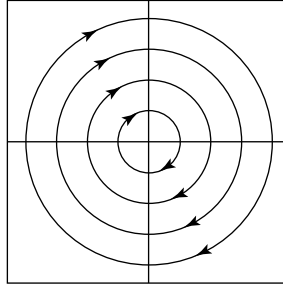


Figure 3.4 Phase portrait for a center.

where c_1 and c_2 are arbitrary constants, provides a solution to any initial value problem.

We claim that this is the general solution of this equation. To prove this, we need to show that these are the only solutions of this equation. So suppose that this is not the case. Let

$$Y(t) = \begin{pmatrix} u(t) \\ v(t) \end{pmatrix}$$

be another solution. Consider the complex function $f(t) = (u(t) + iv(t))e^{i\beta t}$. Differentiating this expression and using the fact that $Y(t)$ is a solution of the equation yields $f'(t) = 0$. Thus, $u(t) + iv(t)$ is a complex constant times $e^{-i\beta t}$. From this it follows directly that $Y(t)$ is a linear combination of $X_{re}(t)$ and $X_{im}(t)$.

Note that each of these solutions is a periodic function with period $2\pi/\beta$. Indeed, the phase portrait shows that all solutions lie on circles centered at the origin. These circles are traversed in the clockwise direction if $\beta > 0$, counterclockwise if $\beta < 0$. See Figure 3.4. This type of system is called a *center*. ■

Example. (Spiral Sink, Spiral Source) More generally, consider $X' = AX$, where

$$A = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}$$

and $\alpha, \beta \neq 0$. The characteristic equation is now $\lambda^2 - 2\alpha\lambda + \alpha^2 + \beta^2$, so the eigenvalues are $\lambda = \alpha \pm i\beta$. An eigenvector associated with $\alpha + i\beta$ is determined by the equation

$$(\alpha - (\alpha + i\beta))x + \beta y = 0.$$

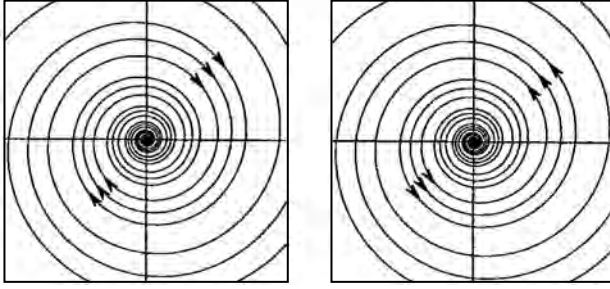


Figure 3.5 Phase portraits for a spiral sink and a spiral source.

Thus $(1, i)$ is again an eigenvector, and so we have complex solutions of the form

$$\begin{aligned} X(t) &= e^{(\alpha+i\beta)t} \begin{pmatrix} 1 \\ i \end{pmatrix} \\ &= e^{\alpha t} \begin{pmatrix} \cos \beta t \\ -\sin \beta t \end{pmatrix} + ie^{\alpha t} \begin{pmatrix} \sin \beta t \\ \cos \beta t \end{pmatrix} \\ &= X_{re}(t) + iX_{im}(t). \end{aligned}$$

As before, both $X_{re}(t)$ and $X_{im}(t)$ yield real solutions of the system with initial conditions that are linearly independent. Thus we find the general solution,

$$X(t) = c_1 e^{\alpha t} \begin{pmatrix} \cos \beta t \\ -\sin \beta t \end{pmatrix} + c_2 e^{\alpha t} \begin{pmatrix} \sin \beta t \\ \cos \beta t \end{pmatrix}.$$

Without the term $e^{\alpha t}$, these solutions would wind periodically around circles centered at the origin. The $e^{\alpha t}$ term converts solutions into spirals that either spiral into the origin (when $\alpha < 0$) or away from the origin ($\alpha > 0$). In these cases the equilibrium point is called a *spiral sink* or *spiral source* respectively. See Figure 3.5. ■

3.3 Repeated Eigenvalues

The only remaining cases occur when A has repeated real eigenvalues. One simple case occurs when A is a diagonal matrix of the form

$$A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}.$$

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The eigenvalues of A are both equal to λ . In this case every nonzero vector is an eigenvector since

$$AV = \lambda V$$

for any $V \in \mathbb{R}^2$. Thus, solutions are of the form

$$X(t) = \alpha e^{\lambda t} V.$$

Each such solution lies on a straight line through $(0,0)$ and either tends to $(0,0)$ (if $\lambda < 0$) or away from $(0,0)$ (if $\lambda > 0$). So this is an easy case.

A more interesting case occurs when

$$A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

Again, both eigenvalues are equal to λ , but now there is only one linearly independent eigenvector that is given by $(1,0)$. Thus, we have one straight-line solution

$$X_1(t) = \alpha e^{\lambda t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

To find other solutions note that the system may be written

$$x' = \lambda x + y$$

$$y' = \lambda y.$$

Thus, if $y \neq 0$, we must have

$$y(t) = \beta e^{\lambda t}.$$

Therefore, the differential equation for $x(t)$ reads

$$x' = \lambda x + \beta e^{\lambda t}.$$

This is a nonautonomous, first-order differential equation for $x(t)$. One might first expect solutions of the form $e^{\lambda t}$, but the nonautonomous term is also in this form. As you perhaps saw in calculus, the best option is to guess a solution of the form

$$x(t) = \alpha e^{\lambda t} + \mu t e^{\lambda t}$$

for some constants α and μ . This technique is often called the *method of undetermined coefficients*. Inserting this guess into the differential equation

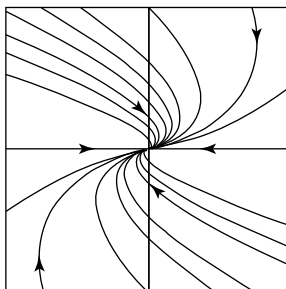


Figure 3.6 Phase portrait for a system with repeated negative eigenvalues.

shows that $\mu = \beta$ while α is arbitrary. Thus, the solution of the system may be written

$$\alpha e^{\lambda t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta e^{\lambda t} \begin{pmatrix} t \\ 1 \end{pmatrix}.$$

This is in fact the general solution (see Exercise 12 of this chapter).

Note that, if $\lambda < 0$, each term in this solution tends to 0 as $t \rightarrow \infty$. This is clear for the $\alpha e^{\lambda t}$ and $\beta e^{\lambda t}$ terms. For the term $\beta t e^{\lambda t}$, this is an immediate consequence of l'Hôpital's rule. Thus, all solutions tend to $(0, 0)$ as $t \rightarrow \infty$. When $\lambda > 0$, all solutions tend away from $(0, 0)$. See Figure 3.6. In fact, solutions tend toward or away from the origin in a direction tangent to the eigenvector $(1, 0)$ (see Exercise 7 at the end of this chapter).

3.4 Changing Coordinates

Despite differences in the associated phase portraits, we really have dealt with only three type of matrices in these past four sections:

$$\begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}, \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}, \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

Any 2×2 matrix that is in one of these three forms is said to be in *canonical form*. Systems in this form may seem rather special, but they are not. Given any linear system $X' = AX$, we can always “change coordinates” so that the new system's coefficient matrix is in canonical form and so is easily solved. Here is how to do this.

A *linear map* (or *linear transformation*) on $\mathbb{R}^2$ is a function $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ of the form

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

That is, T simply multiplies any vector by the 2×2 matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We will thus think of the linear map and its matrix as being interchangeable, so that we also write

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Hopefully no confusion will result from this slight imprecision.

Now suppose that T is *invertible*. This means that the matrix T has an *inverse matrix* S that satisfies $TS = ST = I$ where I is the 2×2 identity matrix. It is traditional to denote the inverse of a matrix T by T^{-1} . As is easily checked, the matrix

$$S = \frac{1}{\det T} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

serves as T^{-1} if $\det T \neq 0$. If $\det T = 0$, we know from Chapter 2 that there are infinitely many vectors (x, y) for which

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus, there is no inverse matrix in this case, for we would need

$$\begin{pmatrix} x \\ y \end{pmatrix} = T^{-1}T\begin{pmatrix} x \\ y \end{pmatrix} = T^{-1}\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

for each such vector. We have shown this.

Proposition. T he 2×2 matrix T is invertible if and only if $\det T \neq 0$. □

Now, instead of considering a linear system $X' = AX$, suppose we consider a different system,

$$Y' = (T^{-1}AT)Y,$$

for some invertible linear map T . Note that if $Y(t)$ is a solution of this new system, then $X(t) = TY(t)$ solves $X' = AX$. Indeed, we have

$$\begin{aligned}(TY(t))' &= TY'(t) \\ &= T(T^{-1}AT)Y(t) \\ &= A(TY(t)),\end{aligned}$$

as required. That is, the linear map T converts solutions of $Y' = (T^{-1}AT)Y$ to solutions of $X' = AX$. Alternatively, T^{-1} takes solutions of $X' = AX$ to solutions of $Y' = (T^{-1}AT)Y$.

We therefore think of T as a change of coordinates that converts a given linear system into one with a different coefficient matrix. What we hope to be able to do is find a linear map T that converts the given system into a system of the form $Y' = (T^{-1}AT)Y$ that is easily solved. And, as you may have guessed, we can always do this by finding a linear map that converts a given linear system to one in canonical form.

Example. (Real Eigenvalues) Suppose the matrix A has two real, distinct eigenvalues λ_1 and λ_2 with associated eigenvectors V_1 and V_2 . Let T be the matrix with columns V_1 and V_2 . Thus, $TE_j = V_j$ for $j = 1, 2$ where the E_j form the standard basis of $\mathbb{R}^2$. Also, $T^{-1}V_j = E_j$. Therefore, we have

$$\begin{aligned}(T^{-1}AT)E_j &= T^{-1}AV_j = T^{-1}(\lambda_j V_j) \\ &= \lambda_j T^{-1}V_j \\ &= \lambda_j E_j.\end{aligned}$$

Thus the matrix $T^{-1}AT$ assumes the canonical form

$$T^{-1}AT = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

and the corresponding system is easy to solve. ■

Example. As a further specific example, suppose

$$A = \begin{pmatrix} -1 & 0 \\ 1 & -2 \end{pmatrix}.$$

The characteristic equation is $\lambda^2 + 3\lambda + 2$, which yields eigenvalues $\lambda = -1$ and $\lambda = -2$. An eigenvector corresponding to $\lambda = -1$ is given by solving

$$(A + I) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which yields an eigenvector $(1, 1)$. Similarly an eigenvector associated with $\lambda = -2$ is given by $(0, 1)$.

We therefore have a pair of straight-line solutions, each tending to the origin as $t \rightarrow \infty$. The straight-line solution corresponding to the weaker eigenvalue lies along the line $y = x$; the straight-line solution corresponding to the stronger eigenvalue lies on the y -axis. All other solutions tend to the origin tangentially to the line $y = x$.

To put this system in canonical form, we choose T to be the matrix with columns that are these eigenvectors:

$$T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

so that

$$T^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Finally, we compute

$$T^{-1}AT = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix},$$

so $T^{-1}AT$ is in canonical form. The general solution of the system $Y' = (T^{-1}AT)Y$ is

$$Y(t) = \alpha e^{-t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta e^{-2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so the general solution of $X' = AX$ is

$$\begin{aligned} TY(t) &= \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \left(\alpha e^{-t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta e^{-2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= \alpha e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta e^{-2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \end{aligned}$$

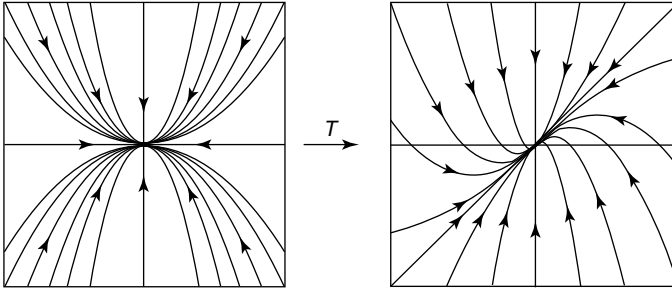


Figure 3.7 Change of variables T in the case of a (real) sink.

Thus the linear map T converts the phase portrait for the system,

$$Y' = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} Y,$$

to that of $X' = AX$ as shown in Figure 3.7. ■

Note that we really do not have to go through the step of converting a specific system to one in canonical form; once we have the eigenvalues and eigenvectors, we can simply write down the general solution. We take this extra step because, when we attempt to classify all possible linear systems, the canonical form of the system will greatly simplify this process.

Example. (Complex Eigenvalues) Now suppose that the matrix A has complex eigenvalues $\alpha \pm i\beta$ with $\beta \neq 0$. Then we may find a complex eigenvector $V_1 + iV_2$ corresponding to $\alpha + i\beta$, where both V_1 and V_2 are real vectors. We claim that V_1 and V_2 are linearly independent vectors in $\mathbb{R}^2$. If this were not the case, then we would have $V_1 = cV_2$ for some $c \in \mathbb{R}$. But then we have

$$A(V_1 + iV_2) = (\alpha + i\beta)(V_1 + iV_2) = (\alpha + i\beta)(c + i)V_2.$$

But we also have

$$A(V_1 + iV_2) = (c + i)AV_2.$$

So we conclude that $AV_2 = (\alpha + i\beta)V_2$. This is a contradiction since the left side is a real vector while the right is complex.

Since $V_1 + iV_2$ is an eigenvector associated with $\alpha + i\beta$, we have

$$A(V_1 + iV_2) = (\alpha + i\beta)(V_1 + iV_2).$$

Equating the real and imaginary components of this vector equation, we find

$$AV_1 = \alpha V_1 - \beta V_2$$

$$AV_2 = \beta V_1 + \alpha V_2.$$

Let T be the matrix with columns V_1 and V_2 . Thus $TE_j = V_j$ for $j = 1, 2$. Now consider $T^{-1}AT$. We have

$$\begin{aligned}(T^{-1}AT)E_1 &= T^{-1}(\alpha V_1 - \beta V_2) \\ &= \alpha E_1 - \beta E_2\end{aligned}$$

and similarly

$$(T^{-1}AT)E_2 = \beta E_1 + \alpha E_2.$$

Thus the matrix $T^{-1}AT$ is in the canonical form

$$T^{-1}AT = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}.$$

We saw that the system $Y' = (T^{-1}AT)Y$ has phase portrait corresponding to a spiral sink, center, or spiral source depending on whether $\alpha < 0$, $\alpha = 0$, or $\alpha > 0$. Therefore, the phase portrait of $X' = AX$ is equivalent to one of these after changing coordinates using T . ■

Example. (Another Harmonic Oscillator) Consider the second-order equation

$$x'' + 4x = 0.$$

This corresponds to an undamped harmonic oscillator with mass 1 and spring constant 4. As a system, we have

$$X' = \begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix} X = AX.$$

The characteristic equation is

$$\lambda^2 + 4 = 0,$$

so that the eigenvalues are $\pm 2i$. A complex eigenvector associated with $\lambda = 2i$ is a solution of the system

$$\begin{aligned} -2ix + y &= 0 \\ -4x - 2iy &= 0. \end{aligned}$$

One such solution is the vector $(1, 2i)$. So we have a complex solution of the form

$$e^{2it} \begin{pmatrix} 1 \\ 2i \end{pmatrix}.$$

Breaking this solution into its real and imaginary parts, we find the general solution

$$X(t) = c_1 \begin{pmatrix} \cos 2t \\ -2 \sin 2t \end{pmatrix} + c_2 \begin{pmatrix} \sin 2t \\ 2 \cos 2t \end{pmatrix}.$$

Thus the position of this oscillator is given by

$$x(t) = c_1 \cos 2t + c_2 \sin 2t,$$

which is a periodic function of period π .

Now, let T be the matrix with columns that are the real and imaginary parts of the eigenvector $(1, 2i)$; that is

$$T = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Then we compute easily that

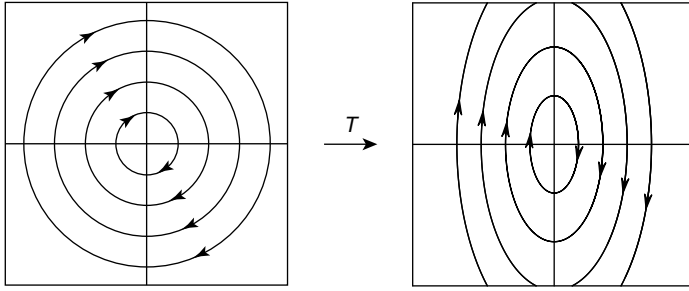
$$T^{-1}AT = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix},$$

which is in canonical form. The phase portraits of these systems are shown in Figure 3.8. Note that T maps the circular solutions of the system $Y' = (T^{-1}AT)Y$ to elliptic solutions of $X' = AX$. ■

Example. (Repeated Eigenvalues) Suppose A has a single real eigenvalue λ . If there exists a pair of linearly independent eigenvectors, then in fact A must be in the form

$$A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix},$$

so the system $X' = AX$ is easily solved (see Exercise 15 of this chapter).

Figure 3.8 Change of variables T in the case of a center.

For the more complicated case, let's assume that V is an eigenvector and that every other eigenvector is a multiple of V . Let W be any vector for which V and W are linearly independent. Then we have

$$AW = \mu V + \nu W$$

for some constants $\mu, \nu \in \mathbb{R}$. Note that $\mu \neq 0$, for otherwise we would have a second linearly independent eigenvector W with eigenvalue ν . We claim that $\nu = \lambda$. If $\nu - \lambda \neq 0$, a computation shows that

$$A\left(W + \left(\frac{\mu}{\nu - \lambda}\right)V\right) = \nu\left(W + \left(\frac{\mu}{\nu - \lambda}\right)V\right).$$

This says that ν is a second eigenvalue different from λ . Thus, we must have $\nu = \lambda$.

Finally, let $U = (1/\mu)W$. Then

$$AU = V + \frac{\lambda}{\mu}W = V + \lambda U.$$

Thus if we define $TE_1 = V$, $TE_2 = U$, we get

$$T^{-1}AT = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix},$$

as required. $X' = AX$ is therefore again in canonical form after this change of coordinates. ■

EXERCISES

1. In Figure 3.9, you see six phase portraits. Match each of these phase portraits with one of the following linear systems:

$$(a) \begin{pmatrix} 3 & 5 \\ -2 & -2 \end{pmatrix} \quad (b) \begin{pmatrix} -3 & -2 \\ 5 & 2 \end{pmatrix} \quad (c) \begin{pmatrix} 3 & -2 \\ 5 & -2 \end{pmatrix}$$

$$(d) \begin{pmatrix} -3 & 5 \\ -2 & 3 \end{pmatrix} \quad (e) \begin{pmatrix} 3 & 5 \\ -2 & -3 \end{pmatrix} \quad (f) \begin{pmatrix} -3 & 5 \\ -2 & 2 \end{pmatrix}$$

2. For each of the following systems of the form $X' = AX$

- Find the eigenvalues and eigenvectors of A .
- Find the matrix T that puts A in canonical form.
- Find the general solution of both $X' = AX$ and $Y' = (T^{-1}AT)Y$.
- Sketch the phase portraits of both systems.

$$(i) A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (ii) A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

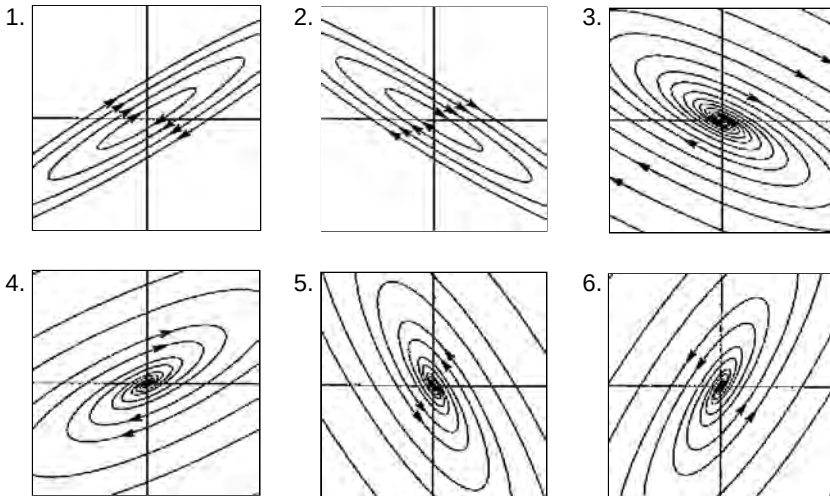


Figure 3.9 Match these phase portraits with the systems in Exercise 1.

$$(iii) A = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \quad (iv) A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}$$

$$(v) A = \begin{pmatrix} 1 & 1 \\ -1 & -3 \end{pmatrix} \quad (vi) A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

3. Find the general solution of the following harmonic oscillator equations:

(a) $x'' + x' + x = 0$

(b) $x'' + 2x' + x = 0$

4. Consider the harmonic oscillator system

$$X' = \begin{pmatrix} 0 & 1 \\ -k & -b \end{pmatrix} X,$$

where $b \geq 0, k > 0$, and the mass $m = 1$.

(a) For which values of k, b does this system have complex eigenvalues? Repeated eigenvalues? Real and distinct eigenvalues?

(b) Find the general solution of this system in each case.

(c) Describe the motion of the mass when the mass is released from the initial position $x = 1$ with zero velocity in each of the cases in part (a).

5. Sketch the phase portrait of $X' = AX$ where

$$A = \begin{pmatrix} a & 1 \\ 2a & 2 \end{pmatrix}.$$

For which values of a do you find a bifurcation? Describe the phase portrait for a -values above and below the bifurcation point.

6. Consider the system

$$X' = \begin{pmatrix} 2a & b \\ b & 0 \end{pmatrix} X.$$

Sketch the regions in the ab -plane where this system has different types of canonical forms.

7. Consider the system

$$X' = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} X$$

with $\lambda \neq 0$. Show that all solutions tend to (respectively, away from) the origin tangentially to the eigenvector $(1, 0)$ when $\lambda < 0$ (respectively, $\lambda > 0$).

8. Find all 2×2 matrices that have pure imaginary eigenvalues. That is, determine conditions on the entries of a matrix that guarantee the matrix has pure imaginary eigenvalues.
9. Determine a computable condition that guarantees that, if a matrix A has complex eigenvalues with nonzero imaginary parts, then solutions of $X' = AX$ travel around the origin in the counterclockwise direction.
10. Consider the system

$$X' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} X,$$

where $a + d \neq 0$ but $ad - bc = 0$. Find the general solution of this system and sketch the phase portrait.

11. Find the general solution and describe completely the phase portrait for

$$X' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} X.$$

12. Prove that

$$\alpha e^{\lambda t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta e^{\lambda t} \begin{pmatrix} t \\ 1 \end{pmatrix}$$

is the general solution of

$$X' = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} X.$$

13. Prove that a 2×2 matrix A always satisfies its own characteristic equation. That is, if $\lambda^2 + \alpha\lambda + \beta = 0$ is the characteristic equation associated with A , then the matrix $A^2 + \alpha A + \beta I$ is the 0-matrix.
14. Suppose the 2×2 matrix A has repeated eigenvalues λ . Let $V \in \mathbb{R}^2$. Using the previous problem, show that either V is an eigenvector for A or else $(A - \lambda I)V$ is an eigenvector for A .
15. Suppose the matrix A has repeated real eigenvalues λ and there exist, a pair of linearly independent eigenvectors associated with A . Prove that

$$A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}.$$

16. Consider the (nonlinear) system

$$\begin{aligned} x' &= |y| \\ y' &= -x. \end{aligned}$$

Use the methods of this chapter to describe the phase portrait.